

SITA RAM MODI

Postdoctoral Research Associate | Experimental & Computational Materials Mechanics

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RESEARCH PROFILE

Materials and mechanical engineering researcher specializing in experimental and computational investigation of fracture, fatigue and failure mechanisms in metallic materials, with particular expertise in Ti-6Al-4V ELI/TC4 ELI alloys. Experience includes tensile, fracture and fatigue testing using an Instron 8801, and Zwick Roell environmentally assisted fracture studies in simulated body fluid, room and elevated temperature SEM/EDS, fractography, electrochemical characterization, and computational fracture modelling using FEM, XFEM and phase-field approaches. Current research focuses on multiphysics fracture modelling and provides a strong foundation for research into environmentally assisted and hydrogen-assisted fracture.

CURRENT & POST-PHD RESEARCH EXPERIENCE

Postdoctoral Research Associate — IIT Mandi, India

School of Mechanical and Materials Engineering | Current

- Working on an AR&DB/DRDO project: development of phase-field modelling and machine-learning-based approaches for fracture analysis of composite structures under multiphysics loadings.
- Developing computational approaches for fracture under coupled/multiphysics conditions, with emphasis on fracture mechanics and numerical implementation.

Research Experience — DRDO/GTRE Project, NIT Warangal, India

- Worked on exploratory studies of fatigue and hold-time effects on crack-growth behaviour of GTM Ti-64 alloy.
- Investigated fatigue crack-growth behaviour with attention to loading conditions, temperature and hold-time effects.

EDUCATION

PhD — IIT (Indian School of Mines), Dhanbad

1 Aug 2019 – 24 Jul 2025 | CGPA: 8.42

Thesis: Failure Mechanism and Life Estimation of Titanium Based Biomedical Hip Implant

- Numerical and experimental crack analysis of annealed TC4 ELI alloy in room and body-fluid environment.
- Comparative investigation of mechanical and electrochemical characteristics of annealed Ti-6Al-4V ELI.
- Life prediction of biomedical implants made of Ti-6Al-4V ELI.
- Multi-mode fracture analysis, critical crack size and life estimation using extended finite element methods.
- Stress-shielding analysis of solid and porous Ti-alloy hip implants using finite-element analysis.

M.Tech — PDPM IITDM, Jabalpur

1 Aug 2017 – 31 Jul 2019 | CGPA: 7.3

Thesis: Finite Element-based Investigation of Ultrasonic Vibration Assisted Incremental Forming Process

B.Tech — Punjab Technical University

1 Aug 2011 – 31 Jul 2015 | 79% | Mechanical Engineering

EXPERIMENTAL MATERIALS ENGINEERING EXPERTISE

- Tensile, fracture and fatigue testing using Instron 8801 and Zwick Roell Universal Testing Machine.
- Fracture toughness and environmentally assisted fracture testing after exposure to simulated body fluid.
- SEM and EDS characterization of as-received and fractured materials.
- Fractography and correlation of fracture-surface features with failure mechanisms.
- Optical microscopy, surface morphology and surface profilometry.
- Electrochemical analysis for corrosion-rate evaluation.
- Vickers microhardness, wear and scratch testing.
- Wettability/contact-angle measurements of metallic alloys.

- Metal additive manufacturing of SS316 and Stellite 21 using Laser Direct Energy Deposition.
- Cladding of biocompatible niobium and vanadium on titanium-alloy substrates using TIG welding deposition.

RESEARCH THEMES & METHODS

- Fracture mechanics, fatigue crack growth, fracture toughness and failure mechanisms.
- Experimental and numerical investigation of Ti-6Al-4V ELI/TC4 ELI alloys.
- Environmentally assisted fracture in simulated body-fluid environments.
- Phase-field fracture, XFEM and finite-element/isogeometric analysis.
- Multiphysics fracture modelling and coupled mechanical/environmental degradation problems.
- MATLAB-based numerical implementation and ABAQUS/ANSYS-based finite-element analysis.

SELECTED PUBLICATIONS

- Modi, S. R., Jha, K. (2023). Multi-mode fracture analysis for critical crack size and life estimation of hip prosthesis using extended finite element method. *Journal of Mechanical Science and Technology*. Best Paper Award, 2023.
- Modi, S. R., Jha, K. (2024). A comparative investigation on mechanical and electrochemical characteristics of annealed Ti-6Al-4V ELI with other grades of titanium alloys. *Journal of Materials Engineering and Performance*.
- Modi, S. R., Ajit, K. P., Jha, K. (2023). Numerical and Experimental Crack Analysis of Annealed TC4 ELI Alloy in the Room and Body Fluid Environment. *Arabian Journal for Science and Engineering*.
- Modi, S. R., Dongare, A., Jha, K. (2025). Strain shielding effect analysis of solid and porous Ti-4Al-6V alloy implanted femur bone using Finite element analysis. *Journal of Medical Engineering & Technology*.
- Modi, S. R., Ajit, K. P., Jha, K. (2026). Strain Amplitude Effects on Fatigue Behavior, Failure Mechanisms, and Life Prediction of Annealed Ti-6Al-4V ELI Alloy. *Fatigue and Fracture of Engineering Materials and Structures*. To be submitted.
- Modi, S. R., Ajit, K. P., Jha, K. (2026). Exploring the Role of Stress Ratios on the Fracture and Fatigue Characteristics of Annealed Ti-6Al-4V ELI Alloy. *International Journal of Fatigue*. To be submitted.
- Modi, S. R., Fernando, C., Srinivasarao, B., Narasaiah, N., Pandey, A. K. (2026). Influence of load ratio and temperature on fatigue crack growth in Ti-6Al-4V alloys for high temperature applications. *International Journal of Fatigue*. To be submitted.

SOFTWARE & COMPUTATIONAL SKILLS

- ABAQUS CAE, ANSYS Workbench, MATLAB, Simufact, CATIA V5, SOLIDWORKS, PTC Creo, AutoCAD, Autodesk Inventor, Fusion 360.
- Finite-element modelling, XFEM, phase-field fracture modelling, stress analysis and numerical fracture simulation.

AWARDS, SERVICE & PROFESSIONAL ACTIVITIES

- Best Paper Award of the year 2023 for multi-mode fracture analysis of hip prosthesis.
- Reviewer for Computational Materials Science, Current Pharmaceutical Biotechnology, Medicon Engineering Themes Journal, Journal of Recent Patents on Engineering, and International Journal of Materials Science and Applications.
- Hands-on workshop: Monitoring & Characterization of Metal Additive Manufacturing (VNIT Nagpur, 2022).

REFEREES

Prof. Ajoy Kumar — Associate Professor, Department of Metallurgical and Materials Engineering, NIT Warangal | ajoy@nitw.ac.in

Prof. Kailash Jha — Associate Professor, Department of Mechanical Engineering, IIT (ISM) Dhanbad | kjha@iitism.ac.in

Prof. Puneet Tandon — Professor of Mechanical Engineering, PDPM IITDM Jabalpur | ptandon@iitdmj.ac.in